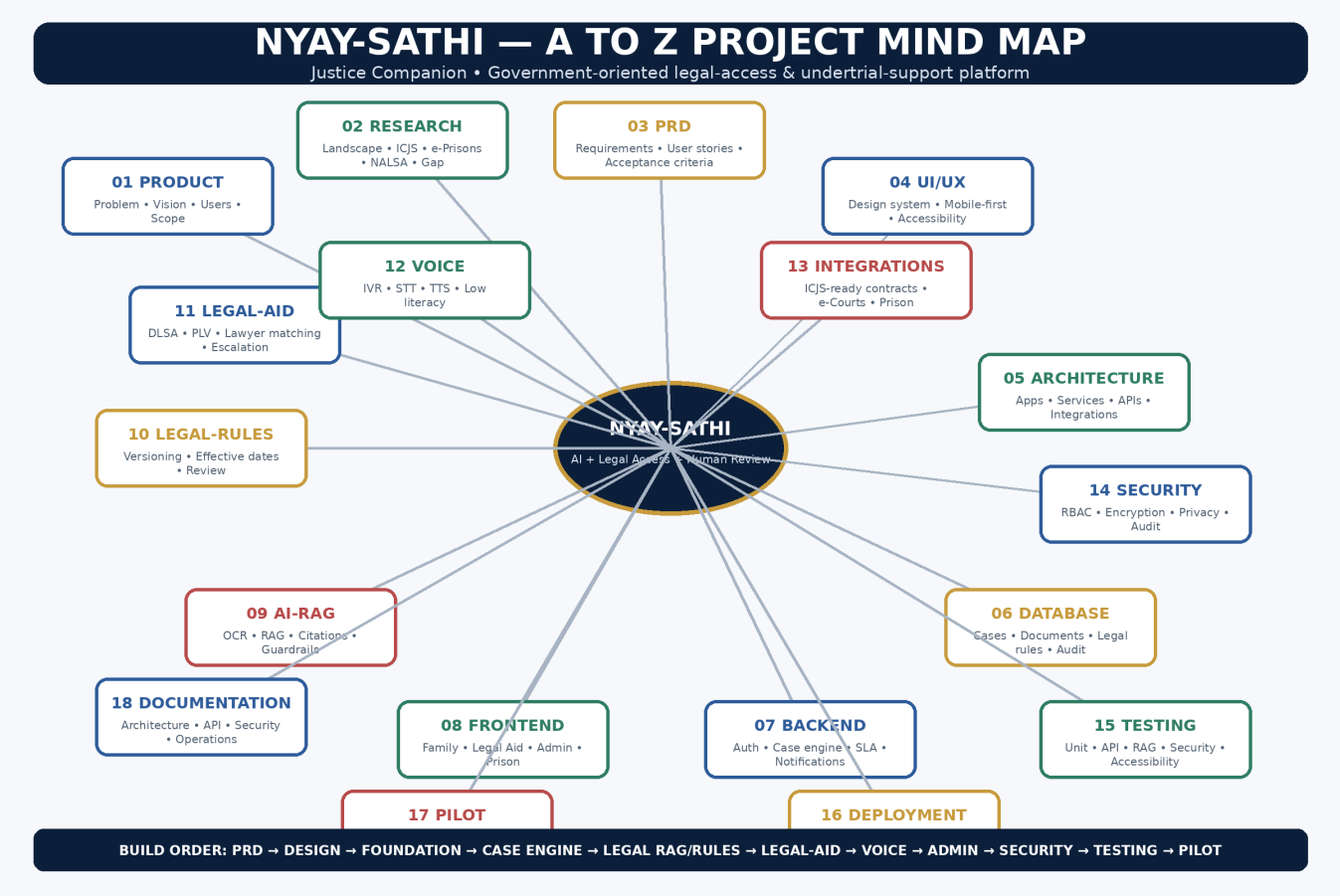
NYAY-SATHI

01 — Government Product Requirements Document (PRD) • Version 1.0 • August 2026

Justice Companion
AI-assisted legal-access and undertrial-support platform for families of undertrial prisoners in India.

Core principle: AI assists. Verified law explains. Humans verify. Courts decide.



Document purpose

This PRD is the master blueprint for taking Nyay-Sathi from concept to a controlled, government-oriented pilot. It defines product scope, users, workflows, architecture, AI/RAG, legal-rule governance, legal-aid routing, security, testing, deployment, pilot governance and measurable outcomes.

Important: This document is a product/engineering specification, not legal advice and not a claim of government endorsement or integration.

1. Executive Summary

Nyay-Sathi is designed as an access, intelligence and coordination layer around India's existing justice and legal-aid ecosystem. It should not replace ICJS, e-Prisons, e-Courts, NALSA/SLSA/DLSA processes, police systems, prisons or courts. The platform's distinctive value is the combination of family-facing access, voice/vernacular support, statute-grounded legal information, potential eligibility screening, human legal review and routing into authorized legal-aid support.

• Family access through web/mobile and voice-oriented channels.

• Case intake and understandable case-status/timeline information.

• Document upload and structured extraction with user/PLV verification.

• Versioned legal-source registry and citation-grounded RAG.

• Rule-based eligibility screening first; ML only after controlled evaluation.

• Legal-aid/lawyer/PLV routing with workload, language, district and SLA considerations.

• Human-override logging, auditability and bias/fairness evaluation.

• Prison-side confirmation channel and URC-ready reporting.

• Rights education, surety/bail-bond navigation and post-release support hooks.

2. Problem Statement

• Families may struggle to understand case status, legal terminology, documents, deadlines and available legal-aid pathways.

• Existing justice infrastructure is fragmented from the perspective of a family member who needs a simple, vernacular, actionable interface.

• Legal information may exist but not be translated into plain-language next steps.

• Potential statutory-review signals can be missed when no one notices or raises the relevant issue.

• Voice/feature-phone access and low-literacy support are central accessibility requirements for the intended user journey.

The user-provided August 2026 analysis identifies ICJS/e-Prisons as internal government infrastructure, e-Courts/NJDG as public case-status portals, research systems such as IBPS/NyayaRAG/NyayaAI, and Nyaya-Sahaya as the closest student-built analogue. It concludes that no identified platfor family-facing, voice-first, statute-grounded and human-routed scope proposed for Nyay-S represented in the source notes below.

Source notes appear in Section 35.

3. Product Positioning

Positioning statement: A digital bridge between families, legal-aid institutions and India's justice ecosystem.

• Not an AI judge.

• Not an autonomous bail-decision system.

• Not a replacement for lawyers.

• Not a replacement for ICJS/e-Prisons/e-Courts.

• An assistive layer for access, understanding, triage, routing, monitoring and accountability.

4. Objectives

• Reduce friction between families and legitimate legal-aid pathways.

• Make case information understandable in simple language and vernacular formats.

• Surface potential legal-review signals with transparent evidence and citations.

• Connect eligible/priority cases to authorized legal-aid professionals.

• Provide proactive, trackable milestones and alerts.

• Create an auditable human-in-the-loop workflow.

• Enable government users to monitor queues, SLAs and unresolved access gaps.

• Prepare the platform for standards-based integration rather than parallel data silos.

5. Target Users & Roles

Role	Primary needs	Access
Family user	Case status, rights, documents, legal aid, updates, voice	Family portal
PLV	Assisted intake, verification, case completion, field support	Legal-aid portal
Legal-aid lawyer	Case queue, documents, AI brief, sources, tasks, review	Legal-aid portal
DLSA/SLSA officer	Triage, allocation, SLA, reports, oversight	Admin/legal-aid portal
Prison officer	Restricted custody/case fact confirmation	Prison channel
Legal reviewer	Verify sources/rules, review AI disagreements	Reviewer portal
State/district admin	Operational analytics and governance	Admin portal
System admin/auditor	Security, audit, configuration	Restricted admin

6. Product Scope — Five Interfaces

• **Family Portal:** landing, onboarding, case search, case dashboard, timeline, documents, legal screening, rights library, legal aid, notifications and voice help.

• **Legal-Aid Portal:** case queue, assignment, AI brief, verified legal sources, tasks, SLA and resolution.

• **Government/Admin Portal:** district/state analytics, queue management, allocation, escalation, URC-ready export and AI audit.

• **Prison-side Channel:** restricted confirmation of custody, transfer, arrest and release-related facts.

• **Voice/IVR Layer:** feature-phone-friendly intake, case status, rights information and connection to support.

7. End-to-End User Journey

Discovery → Intake → Case creation → Document/record verification → Legal-source retrieval → Rule screening → Explainable result → Human legal review → Legal-aid routing → Family notification → Hearing/action → Resolution → Post-release follow-up.

• Allow incomplete intake and PLV-assisted completion when the family does not know an FIR/CNR number.

• Show source, date and verification status for important case/legal information.

• Every high-impact AI signal must provide a human-review path.

• Every important workflow action must generate a case event and audit record.

8. Family Portal Requirements

• Home: Search Case, Check Legal Screening, Upload Document, Get Legal Aid, Know Your Rights, Voice Help.

• Case dashboard: case identifier, status, custody information when verified, next action, legal-aid status and timeline.

• Document center: upload, preview, extraction, field verification and secure access.

• Rights library: simple-language cards, audio, language selection and source metadata.

• Notifications: case updates, document requests, lawyer assignment, hearing reminders and configured escalation alerts.

• Accessibility: mobile-first, large touch targets, screen-reader semantics, reduced motion and low-bandwidth states.

9. Legal-Aid Workflow

Family/PLV request → eligibility/priority triage → district/language/case-type matching → lawyer/PLV assignment → SLA timer → human review → action → resolution.

• Maintain verified lawyer/PLV profiles and availability.

• Never fabricate or display an unverified professional as verified.

• Use configurable SLA policies rather than hard-coded assumptions.

• Support escalation to supervisors when configured response windows are approaching or breached.

10. Case & Timeline Engine

• Case states: NEW, INTAKE_INCOMPLETE, UNDER_REVIEW, LEGAL_AID_REQUESTED, LAWYER_ASSIGNED, ACTION_REQUIRED, HEARING_SCHEDULED, RESOLVED, ESCALATED.

• Events: case created, document uploaded, extraction verified, screening completed, human review, lawyer assigned, hearing scheduled, outcome recorded, release confirmed.

• Every event stores actor, timestamp, source, status and relevant evidence/reference.

11. Document Intelligence

PDF/JPG/PNG → OCR → structured extraction → confidence per field → source page → user/PLV verification → case record.

• Extract FIR number, police station, accused name, sections, dates, court and other supported fields.

• Never silently trust OCR.

• Store extraction confidence and source location.

• Allow correction and retain an audit trail.

12. Legal Knowledge & RAG

• Create an approved legal-source registry.

• Store law/section/title/version/effective dates/source URL/verification status/reviewer.

• Chunk and index approved documents.

• Retrieve relevant provisions before generation.

• Require citation-backed substantive legal explanations.

• Run citation validation and safety checks before returning answers.

• If no approved source supports the answer, the system must abstain and route to human support.

RAG rule: Retrieval is a prerequisite for legal explanation; free-form model knowledge must not be treated as authoritative legal source material.

13. Eligibility / Screening Engine

The first production-oriented implementation should be a versioned, rule-based screening engine. ML should be a later, separately evaluated layer.

• Normalize case facts.

• Load applicable rule versions.

• Evaluate conditions and exceptions.

• Return a potential signal, not a court outcome.

• Show detected facts, rule/source, missing information and confidence band.

• Require human verification for consequential actions.

• Keep legal rules separate from UI code.

14. Confidence & Explainability

• Use High/Medium/Low or equivalent bands for screening confidence.

• Define confidence as information/model reliability, not probability of a court granting bail.

• Show why a signal was raised.

• Show which facts are missing or unverified.

• Show exact legal source and version.

• Record reviewer agreement/disagreement.

15. Human-in-the-Loop

AI signal → legal reviewer/PLV/lawyer → agree or override → reason → audit log → evaluation dataset.

• Every override requires a reason.

• Never hide AI-human disagreement.

• Use disagreements for audit and model/rule improvement only under approved governance.

• AI must remain advisory.

16. Bias & AI Evaluation

• Build a held-out evaluation set before trusting an ML layer.

• Measure retrieval accuracy, citation correctness, hallucination rate, rule accuracy and human agreement.

• Track false positives and false negatives.

• Audit performance across operational dimensions such as district, offence category, language and legal-aid availability where appropriate.

• Have legal/ethics reviewers inspect disagreements before scale.

17. New / Differentiating Modules

• Family-side literacy nudges: short contextual rights explanations.

• Surety/bail-bond navigator using verified support resources.

• Prison-side confirmation channel.

• Post-release reintegration hooks.

• URC-ready export and triage support.

• Legal-source change management.

• Document comparison / 'what changed?' feature.

• Action-oriented 'What do I do next?' guidance.

18. Integration Strategy

Nyay-Sathi should use an adapter architecture: Core Case Service → Integration Interface → ICJS/e-Courts/prison/legal-aid connectors when official access is authorized.

• Do not build unofficial scrapers for government systems.

• Use documented APIs, approved data-sharing mechanisms, or manual verified entry where official integration is unavailable.

• Define stable CaseRecord, CustodyRecord, CourtRecord, LegalAidRequest and CaseEvent contracts.

• Keep integration failures visible and auditable.

19. Data Model — Core Entities

• Identity: users, roles, permissions, families, prisoners.

• Case: cases, case_events, case_statuses, case_priorities.

• Documents: documents, document_pages, document_extractions.

• Legal knowledge: legal_sources, legal_rules, legal_rule_versions, legal_citations.

• AI: eligibility_checks, eligibility_results, model_versions, model_evaluations, bias_audits.

• Legal aid: lawyers, lawyer_languages, lawyer_specializations, legal_aid_requests, lawyer_assignments, sla_records.

• Operations: notifications, voice_sessions, prison_confirmations, human_reviews, human_overrides, audit_logs.

• Integration: organizations, districts, police_stations, courts, prisons, integration_records.

20. API Domains

```
/auth • /users • /cases • /documents • /legal • /eligibility • /legal-aid • /lawyers • /assignments • /notifications • /voice • /prison • /admin • /audit • /analytics • /integrations
```

21. Security & Privacy

• RBAC and least privilege.

• Private document storage with signed access.

• Encryption in transit and at rest.

• Secure authentication/session management.

• Input validation and rate limiting.

• Malware scanning for uploaded files.

• Audit logs for sensitive access and changes.

• Backups and disaster recovery.

• Development uses synthetic/composite data only.

• Production privacy, retention, consent/notice and data-sharing rules require formal legal/security review.

22. Voice / IVR

• Feature-phone-friendly call flow.

• Language selection.

• Speech-to-text and text-to-speech architecture.

• Same backend case and legal services as web.

• Low-literacy prompts.

• Fallback to PLV/support when confidence is low.

• Do not make voice responses sound like final legal advice.

23. UX / Design System

• Premium civic-tech visual language: trustworthy, calm, accessible, human.

• Mobile-first with large touch targets.

• Inter + Noto Sans Tamil (or equivalent multilingual font strategy).

• Subtle motion; respect prefers-reduced-motion.

• Use progressive disclosure to avoid overwhelming families.

• Always show status, next action and source where relevant.

24. Technology Recommendation

Layer	Recommended starting stack
Frontend	React + TypeScript + Vite + Tailwind CSS + shadcn/ui + Framer Motion
Backend	Node.js + Express + TypeScript
Database	PostgreSQL / Supabase
Vector search	pgvector
Storage	Private object storage / Supabase Storage during prototype
AI	Provider-agnostic LLM API behind backend
OCR	Provider-agnostic OCR service
Charts	Recharts
Development deployment	Vercel + Render/Railway + Supabase
Production	Government-approved infrastructure and security architecture

25. Repository / Workspace Structure

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NYAY-SATHI/
■■■ 01-PRODUCT/
■■■ 02-RESEARCH/
■■■ 03-PRD/
■■■ 04-UI-UX/
■■■ 05-ARCHITECTURE/
■■■ 06-DATABASE/
■■■ 07-BACKEND/
■■■ 08-FRONTEND/
■■■ 09-AI-RAG/
■■■ 10-LEGAL-RULES/
■■■ 11-LEGAL-AID/
■■■ 12-VOICE/
■■■ 13-INTEGRATIONS/
■■■ 14-SECURITY/
■■■ 15-TESTING/
■■■ 16-DEPLOYMENT/
■■■ 17-GOVERNMENT-PILOT/
■■■ 18-DOCUMENTATION/

Implementation workspace:
apps/ services/ packages/ database/ legal/ ai/ integrations/ tests/ docs/ scripts/
```

26. A-to-Z Implementation Plan

Phase	Deliverables
0 — Governance & research	Stakeholders, legal review, existing-system map, scope, risks.
1 — PRD & acceptance criteria	Freeze requirements, roles, user stories, non-functional requirements.
2 — UI/UX	Figma/design system, mobile-first flows, accessibility, prototype testing.
3 — Foundation	Workspace, repo, auth, RBAC, database, API skeleton, logging.
4 — Case engine	Case creation/search, timeline, documents, notifications.
5 — Document AI	Upload, OCR, extraction, verification.
6 — Legal knowledge	Approved source registry, versioning, indexing, retrieval.
7 — Rule engine	Versioned legal rules, screening signals, explainability, human review.
8 — Legal RAG	Retrieval, citations, abstention, guardrails, evaluation.
9 — Legal-aid routing	Lawyer/PLV registry, matching, assignment, SLA, escalation.
10 — Voice	IVR-ready flow, STT/TTS, language, fallback.
11 — Admin/government	Dashboards, analytics, URC-ready export, audit center.
12 — Prison channel	Restricted custody/transfer/release confirmation.
13 — Security	Threat model, access control, secure files, privacy, audit.
14 — Testing	Unit, integration, API, RAG, rule, accessibility, performance, security.
15 — Pilot	Single district/state controlled pilot, training, governance, KPI measurement.
16 — Scale readiness	Interoperability, operations, support, disaster recovery, procurement/governance.

27. Antigravity + Claude Operating Method

• **Antigravity:** primary implementation agent/workspace. Open the project folder as a workspace, keep review-driven development enabled, use implementation plans/tasks/walkthroughs, and work phase-by-phase.

• **Claude:** senior reviewer for architecture, code review, security review, RAG prompt review, documentation and adversarial testing.

• **Rule:** no agent may silently change the architecture or legal assumptions; update the PRD/decision log first.

• **Evidence:** after every phase, keep tests, screenshots/walkthroughs, decision notes and a phase status record.

28. Antigravity Workspace Setup

• Create one empty local folder named NYAY-SATHI.

• Open that folder in Antigravity as the workspace; Antigravity's official codelab describes Open Folder as the starting point for a local workspace.

• Use the Agent panel to request planning before coding.

• Use review-driven development and inspect terminal/file-system permissions before allowing changes.

• Use workspace Rules under .agents/rules/ and Workflows under .agents/workflows/ to preserve project standards.

• Keep the project skeleton and PRD inside the workspace so the agent always has the same source of truth.

Official Antigravity guidance: https://codelabs.developers.google.com/getting-started-agy-ide

29. Master Source-of-Truth Files

• 03-PRD/01_NYAY_SATHI_GOVERNMENT_PRD.pdf — product requirements.

• 05-ARCHITECTURE/SYSTEM_ARCHITECTURE.md — current architecture.

• 10-LEGAL-RULES/LEGAL_RULE_GOVERNANCE.md — legal-source/rule governance.

• 14-SECURITY/SECURITY_BASELINE.md — security controls.

• 17-GOVERNMENT-PILOT/PILOT_PLAN.md — pilot scope and governance.

• 18-DOCUMENTATION/PROJECT_STATUS.md — phase status and blockers.

• .agents/rules/ — persistent Antigravity workspace rules.

• .agents/workflows/ — reusable build/test/review workflows.

30. Acceptance Criteria

• Every major user journey works end-to-end using synthetic data.

• Family can create/find a case and understand its current status.

• Documents can be uploaded, extracted and verified.

• Legal explanations cite approved, versioned sources or abstain.

• Eligibility screening produces explainable advisory signals.

• Human review and override are auditable.

• Legal-aid matching and SLA workflow works.

• Admin dashboards show operational metrics.

• Voice architecture has a tested demo/fallback.

• Security and authorization tests pass.

• Mobile and accessibility tests pass.

• Pilot plan defines stakeholders, training, governance, support and KPIs.

31. KPIs / Success Metrics

• Median time from legal-aid request to verified assignment.

• Percentage of cases successfully routed.

• Percentage of substantive legal responses with valid citations.

• Document extraction accuracy on a labelled test set.

• AI-human disagreement/override rate.

• False-positive and false-negative rates for any evaluated ML component.

• SLA breach rate.

• Notification delivery success rate.

• Family comprehension / task-completion rate.

• Voice interaction completion rate.

• System availability and incident response metrics.

32. Risk Register

Risk	Impact	Mitigation
Legal hallucination	Critical	Approved-source RAG, citation validation, abstention, human review
Outdated law	Critical	Versioned legal sources, effective dates, legal reviewer
AI bias	High	Held-out evaluation, disaggregated audit, human override
Unauthorized access	Critical	RBAC, least privilege, secure storage, audit
Government integration unavailable	High	Adapter architecture + manual verified fallback
Poor OCR	Medium/High	Confidence per field + user/PLV verification
Low digital literacy	High	Voice, simple language, PLV support
Over-automation	High	Advisory-only positioning, human review
Operational adoption	High	Pilot, training, SOP alignment, stakeholder co-design

33. Pilot Strategy

• Start with one state and a tightly scoped district/pilot environment.

• Prefer a legal-aid institution as the first operational stakeholder because the platform's immediate value is access and routing.

• Use synthetic data during development and controlled, approved data only during a formal pilot.

• Define governance, legal review, security review, user training, support and incident escalation before live use.

• Do not claim government partnership, official integration or endorsement until formally obtained.

34. Two Core Demonstration Journeys

Journey A — Family access: feature-phone/portal intake → incomplete case → PLV completion → document verification → screening → plain-language explanation → legal-aid routing → lawyer assignment → family notification → hearing/action → resolution.

Journey B — Urgent statutory review: custody/investigation data → versioned rule engine → potential review signal → urgent priority → DLSA/legal-aid queue → human verification → action → audit.

35. Research Basis & Source Notes

• User-provided NYAY-SATHI Project Analysis Report, August 2026: existing-solutions landscape, feature comparison, recommended additions, illustrative Ramesh/Iqbal case studies and pilot recommendation.

• Ministry of Home Affairs — ICJS: https://www.mha.gov.in/en/commoncontent/inter-operable-criminal-justice-system-icjs

• NALSA — FAQs / legal-aid information: https://nalsa.gov.in/faqs/

• NALSA — Prisoner/legal-aid SOP materials: https://chandigarh.nalsa.gov.in/

• TNSLSA — Legal Aid: https://tamilnadu.nalsa.gov.in/legal-aid/

• Google Antigravity official codelab: https://codelabs.developers.google.com/getting-started-agy-ide

Source-derived basis from the user-provided report: the landscape comparison states that ICJS/e-Prisons provide government infrastructure, research systems cover legal RAG/bail prediction, Nyaya-Sahaya is the closest student project, and the proposed

Source basis: user-provided NYAY-SATHI Analysis Report, August 2026; feature comparison and landscape findings.

The same report recommends citation-backed rationale, confidence banding, held-out evaluation, ICJS-native contracts, URC export, bias

36. Final Product Principle

NYAY-SATHI should be built as a trustworthy civic-tech system where **AI detects, verified law explains, humans verify, and the justice system decides.** The project's credibility comes from interoperability, legal-source governance, human control, accessibility, security and measurable operational improvement—not from claiming that an AI can predict or decide court outcomes.